AI Won’t Save or Ruin School, But It Can Quietly Make Students Learn Less

If you’re a student right now, you’ve probably used a generative AI tool at least once. If you’re a teacher, you’ve probably marked work that feels... oddly polished. And if you’re a parent, you’ve probably heard both extremes: *AI will finally personalise learning for every child*, or *AI will destroy education*.

As an undergraduate psychology student, I can’t help noticing that learning isn’t the same thing as producing a good-looking answer. Learning is what happens when a student struggles *just enough*, checks their understanding, corrects errors, and tries again. That “mental workout” is exactly what a good tool should support, not replace.

So when I hear people say, “It’s either AI everywhere or a total ban,” I hear the classic either/or trap. In critical thinking terms, that’s a false dichotomy: two dramatic options presented as if there’s no workable middle ground (Halpern & Dunn, 2023). And right now, while schools are writing rules and regulators are setting boundaries, it matters that we get the middle ground right (European Commission, n.d.).

The evidence since 2023 points to a surprisingly practical conclusion: generative AI can improve learning when it’s designed like a structured tutor that keeps students thinking, but it can undermine learning when it becomes a shortcut that does the thinking for them (Kestin et al., 2025; OECD, 2026). I’ll call these two patterns coach mode and crutch mode.

My position is hybrid and conditional: use AI in education, but design and govern it so it stays in coach mode. That means nudging planning, self-explanation, checking, and revision, while blocking “instant solution” use where it predictably weakens learning.

The hype and the panic are both selling you a story

Here’s a common “pro-AI” pitch you’ve probably seen: a slick demo where a chatbot explains a topic beautifully and writes a perfect paragraph. It’s compelling, because our brains love fluent, confident output. But if someone shows you only the best-case examples and calls that “the evidence,” that’s not careful reasoning. It’s selective evidence what Halpern describes as card stacking (Halpern & Dunn, 2023).

On the other side, you’ll hear: “If students use AI for homework, soon they won’t be able to think at all.” That’s a familiar fear pattern too: one step becomes an inevitable collapse. That’s essentially a slippery slope argument and it often skips the key question: *under what conditions does the harm happen, and under what conditions does it not?* (Halpern & Dunn, 2023).

A better question is boring but useful: *When does AI help students learn, and when does it help them avoid learning?*

When AI helps, it doesn't act like a chatbot. It acts like a tutor with rules.

Some of the strongest evidence comes from studies where AI is used in a structured, tutor-like way.

In one randomized controlled trial in a U.S. university, students using an AI tutoring system learned more than students in an active-learning class (Kestin et al., 2025). The details are technical, but the takeaway is simple: this wasn't "ask anything and get an answer." It was guided practice and feedback closer to tutoring than to text generation.

The global equity angle matters too. A World Bank randomized trial in Nigeria tested a six-week after-school program combining teacher facilitation with Microsoft Copilot (GPT-4). Students showed meaningful improvements on curriculum-aligned outcomes, including English, and the authors emphasised the results depended on implementation quality and appropriate use conditions (De Simone et al., 2025). In plain terms: AI didn't magically fix learning. A well-run learning environment used AI to amplify learning.

A meta-analysis of experimental studies also found that ChatGPT-based interventions tended to improve academic performance and motivation overall (Deng et al., 2025). But there's a twist in the same results: students also reported lower mental effort (Deng et al., 2025). That can be good (scaffolding) or bad (offloading). Which it is depends on whether the tool supports thinking or replaces it.

When AI harms, it's usually not because students are "bad." It's because humans take shortcuts.

This is where psychology helps. People are not broken for wanting the easiest route. We all do it. If a tool makes a task easier, most people will use it to reduce effort, especially under time pressure.

The OECD's *Digital Education Outlook 2026* summarises a study with nearly 1,000 high-school maths students that shows exactly why this matters. With unrestricted GPT access, students performed much better during practice, but then did worse on an exam when the tool was removed. The OECD calls this the "Crutch Effect": the tool boosts performance in the moment while weakening independent mastery later (OECD, 2026).

That pattern should ring alarm bells for educators because it reveals a common illusion: practice can look great while learning is quietly shrinking. In psychology, we know fluency can be mistaken for understanding. AI makes that confusion easier, because it produces answers that feel confident, even when the student hasn't built the skill.

A school-based trial in Italy shows how quickly a "tutor" can drift into "answer machine." Students generally liked GPT-4 as a homework tutor, but overall learning gains were mixed, and the researchers found that the tutor sometimes revealed answers directly rather than guiding problem-solving (Vanzo et al., 2024). When "help" becomes "here's the solution," the student's job changes from learning to copying, even if nobody intends it.

The real battleground is something most AI debates barely mention: self-control while studying

Tech debates often focus on *what the model can do*. Psychology keeps asking: what does the tool train the student to do?

Think about studying as having a small coach in your head. That coach asks: *Do I actually get this? What's confusing me? What should I try next?* That is self-regulated learning.

Researchers looking at how students use educational chatbots repeatedly find a creeping issue: the more the tool decides, the less the student does, especially for the hard parts like checking work, reflecting on mistakes, and adjusting strategy (Guan et al., 2025; Lan & Zhou, 2025). In human terms, it's the difference between:

- "Give me a hint so I can solve it," and
- "Just tell me what to write."

The first keeps the student in charge. The second hands the steering wheel to the tool.

There's also a separate issue: using generative AI responsibly requires *good judgment*. You have to notice when output is wrong, incomplete, or misleading. That means monitoring and verification metacognition. Research on generative AI use emphasises that these metacognitive demands are real: users need to plan, evaluate, and calibrate trust, not simply accept output because it sounds confident (Tankelevitch et al., 2023).

And here's where you can spot a bias in everyday life. Picture a student saying, "It wrote it so confidently, so it must be right." That's not laziness; it's automation bias over-trusting an automated system (Halpern & Dunn, 2023; Tankelevitch et al., 2023). Once you recognise that pattern, you'll see it everywhere, not just with AI, but with GPS directions, spellcheck, and "smart" recommendations.

What schools can do (without pretending AI will disappear)

This doesn't need to be a culture war. We already have enough evidence to justify a simple, practical direction: design for coach mode, block crutch mode, and teach students how to check their own thinking.

Here are five changes that fit what the evidence keeps pointing to:

1) Make "attempt first" the default.
Coach-mode AI works best when students try before they ask. Otherwise the tool becomes the first step, not the support step. This aligns with the OECD's warning about crutch effects under unrestricted access (OECD, 2026).

2) Build assessments that reward thinking, not polish.
If grading rewards perfect output, AI will be used to generate perfect output. If grading rewards explanation, transfer, and reasoning, AI becomes less useful as a shortcut and more useful as a coach (OECD, 2026).

3) Teach a three-step verification habit.
Not a vague "be careful," but a routine:

- *What is the claim?*
- *What evidence supports it?*
- *How could it be wrong?*

  That kind of disconfirming check is exactly the "transfer" goal in critical thinking: recognising persuasion patterns and testing conclusions in real contexts (Halpern & Dunn, 2023).

4) Treat educational AI like a safety-critical system, not a fun app.
The U.S. Department of Education argues for guardrails grounded in equity, privacy, and transparency (U.S. Department of Education, 2023). NIST's AI Risk Management Framework and its generative AI profile spell out concrete expectations testing, documentation, monitoring, and mitigation rather than vague reassurance (NIST, 2023, 2024). UNESCO similarly stresses privacy protections and age-appropriate safeguards in education (UNESCO, 2023). This is the boring part, but it's the part that prevents harm.

5) Don't let "everyone's doing it" make the decision for you.
You'll hear: "All students use it anyway." That's basically bandwagon persuasion: popularity presented as proof (Halpern & Dunn, 2023). The right response isn't denial, it's design. If students are using AI, schools need to shape *how* it's used.

Where I land: stop arguing about AI in general, and start arguing about coach vs crutch

If we keep debating "AI: yes or no," we'll keep swinging between hype and panic. Critical thinking pushes us to a better question: what would have to be true for AI to increase learning, not just output?

The evidence so far gives a clear answer. AI helps when it keeps the student in charge prompting planning, self-explanation, checking, and revision. AI harms when it quietly replaces those processes.

So no, AI won't automatically save or ruin school. But it can absolutely make learning better, or quietly make it thinner. The difference is whether we build it, teach it, and govern it as a coach, not a crutch.

Methods

I applied critical thinking as purposeful, reasoned, goal-directed thinking that uses evidence and aims to overcome bias (Halpern & Dunn, 2023). Instead of treating the issue as a binary "for/against" debate, I identified common persuasion patterns in public discussion and then showed how they operate in everyday examples before naming them (transfer): either/or framing (false dichotomy), selective demos presented as proof (card stacking), popularity treated as evidence (bandwagon persuasion), and "one step means catastrophe" reasoning (slippery slope) (Halpern & Dunn, 2023). To reduce motivated reasoning and confirmation bias, I intentionally included strong evidence supporting benefits (structured tutoring RCTs) and strong evidence showing risks (the OECD "Crutch Effect" under unrestricted access), then focused on the conditions that explain divergence (Halpern & Dunn, 2023; OECD, 2026). Finally, I used Toulmin's model implicitly to keep claims appropriately qualified: the claim is conditional ("AI helps when designed as coach mode"), grounded in recent empirical studies, supported by a warrant from learning psychology (durable learning depends on effortful practice plus metacognitive monitoring), and bounded by rebuttals ("unless" AI reveals answers or incentives reward output over reasoning) (Toulmin, 1958).

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